

Anushka Dwivedi

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EDUCATION

University of Wisconsin-Madison | Madison, WI

May 2026

Bachelor of Science: Computer Science and Data Science | Dean's List (3x) | GPA: 3.6

- Certificate in Business | Organizations and Involvements: **Women in Computing (WACM)**, **Alpha Omega Epsilon Sorority**
- Relevant Coursework: Artificial Intelligence, Algorithms, Operating Systems, Computer Networks, Big Data Systems, Linear Algebra, Database Management Systems, Machine Organization, Causal Inference

SKILLS

Languages: Python, Java, TypeScript, JavaScript, C++, SQL, C

Frameworks: React, FastAPI, Node.js, Express, Angular, .NET, Playwright, LangGraph, Pydantic, Zod

AI/ML: LangChain, RAG, FAISS, pgvector, OpenAI API, Gemini API, Llama, PyTorch, Scikit-learn

Data/Systems: PostgreSQL, SQLite, MongoDB, Cassandra, Kafka, Hadoop, Docker, Linux, CMake, GCP, Git, GitHub Actions, CI/CD

PROFESSIONAL EXPERIENCE

PMT College | **Software Development Engineer Intern** | Lucknow, Uttar Pradesh

June 2025 – August 2025

- Built a **TypeScript and Angular** testing platform for **300+ users** with timed assessments and question navigation.
- Integrated answer persistence and submission workflows with **.NET APIs**, preserving test state and validating responses.
- Reduced API **latency by 25%** by tracing backend requests, isolating bottlenecks, and deploying production fixes.

University of Wisconsin-Madison | **Undergraduate Student Researcher** | Madison, Wisconsin

September 2025 – May 2026

- Built a **10+ module Python, LangChain, and Llama pipeline** to extract learning objectives from course materials.
- Improved classification **accuracy by 80%** through prompt optimization and systematic evaluation of input configurations.

Center of Academic Excellence | **Academic Peer Mentor** | Madison, Wisconsin

September 2025 – May 2026

- Mentored CS and math students in **recursion, OOP, and algorithms**, adapting instruction to individual learning needs.

PROJECTS

CodePilot

- Built a **repository-level coding agent** using **Python, LangGraph, and FAISS**, orchestrating a **6-stage workflow** for codebase analysis, semantic retrieval, planning, patch generation, validation, and debugging.
- Engineered **FAISS-based semantic search**, structured code editing, path-traversal protection, dry-run patch review, and **human-in-the-loop approval** to safely modify multi-file repositories.
- Implemented bounded **self-repair** and automated testing with **35 passing unit tests**, demonstrating recovery from failed validation in **1 repair iteration** and successful completion of a **12-test multi-file change**.

Resonance

- Built an explainable music discovery engine that interprets natural-language moods, contexts, and lyrical narratives across **89,740 songs** with **100% sound-embedding coverage**.
- Designed a hybrid recommendation pipeline combining **3 retrieval channels** and **8 relevance signals**; redesigned its embedding representation to reduce input tokens by **41%**, from **9.80M to 5.74M**.
- Developed structured LLM intent parsing, resumable cost-controlled data pipelines, **7 database migrations**, and **54 automated backend tests**.

Clinical Trial Immune Cell Analytics Dashboard

- Built a full-stack clinical analytics dashboard using **React, TypeScript, FastAPI**, and **SQLite** to process **10,500 patient samples** and **52,500 immune-cell measurements**.
- Implemented data validation, **14 unit and integration tests**, **GitHub Actions CI**, and **Vercel deployment**, creating a reproducible workflow with one-command setup, analysis, and dashboard execution.

Resilient Market Data Engine

- Built a **multithreaded C++20 market data pipeline** that validated binary UDP messages and maintained a **price time priority order book**
- Implemented **sequence gap detection, replay, and snapshot recovery**, verifying exact state reconstruction across **50,000 randomized events** with injected message loss and reordering
- Benchmarked order book event application at **7.7M events per second** with **209 ns p99 measured latency** across a reproducible **200,000 event** Release workload